

RotorLab — Sliding Mesh Setup Log

Point-by-point record of every setting changed to convert the converged steady-MRF case into the transient sliding-mesh case, and the reasoning behind each one. Companion to the Build & Decision Log (geometry/mesh) and MRF Results Report (steady-phase results).

0. Context

Starting point: RotorLab(Sliding Mesh).sim, a Save-As copy of the fully converged steady MRF case (1500 iterations, torque settled at -0.41666 N·m). Same geometry and mesh throughout — only Motion, Interface, Continuum, and solver/monitor settings change. Ten settings were touched, in the order below (not the order first listed, since two of them turned out to have a dependency on each other — see Section 1).

1. Region Motion: MRF Reference Frame → Real Rotation

Item	Before (MRF)	After (Sliding Mesh)
RotatingFluid > Motion	Stationary	Rotation (Tools > Motions > Rotation)
RotatingFluid > Reference Frame	ReferenceFrame for Rotation	Lab Reference Frame
Rotation object Rate/Axis/Origin	31.416 rad/s, [0,0,1], [0,0,0]	unchanged, reused

Why: MRF is a mathematical trick — the mesh never physically moves, only the reference frame the equations are solved in rotates, which is why Reference Frame carried the rotation before. Sliding mesh needs the mesh itself to physically move, which requires a true rigid-body Motion object assigned directly to the region, with the frame reverted to Lab (otherwise the frame and the mesh would both be rotating, doubling the effect).

Snag hit: the Rotation motion object already existed under Tools > Motions, but did not appear as a selectable option in RotatingFluid's Motion dropdown. Root cause: a Region's Motion dropdown only offers true rigid-body motions once the Physics Continuum's time model is Unsteady — under Steady (still set from the MRF phase), there is no time axis for the mesh to move through, so STAR-CCM+ correctly hid it as invalid. Fix: do Section 3 (switch continuum to Implicit Unsteady) before this step, not after.

2. Interface: Enable Reset on Relative Motion

Interfaces → RotatingFluid/StationaryFluid → **Reset on Relative Motion**: changed from unchecked to **checked**. Type (Internal Interface), Topology (In-place), and Connectivity (Imprinted) were left unchanged.

Why: with Reset on Relative Motion off, STAR-CCM+ computes the flux mapping between the two sides once and treats it as fixed — correct for MRF, where nothing physically moves. With the mesh now physically rotating, that mapping has to be recomputed every timestep as the rotor sweeps past the stationary baffles, which is exactly what this flag enables.

Confirmed working (not just assumed): once the run started, the output log's "Updating 1 boundary mode interface" line showed the intersection face count and adapted-side area percentage genuinely changing timestep to timestep (e.g. 42347 faces/78.458% at TimeStep 1, 42476/78.4292% at TimeStep 2, continuing to shift at every step thereafter) — direct, mechanistic proof the interface is recalculating as the rotor moves, not a cosmetic setting.

3. Physics Continuum: Steady → Implicit Unsteady

Continua > Physics 1 > Models: time model changed from Steady to **Implicit Unsteady**, Temporal Discretization left at 1st-order. Realizable K-Epsilon Two-Layer, Two-Layer All Y+ wall treatment, segregated flow, and liquid water (998 kg/m³, 0.001 Pa·s) all left unchanged. **Why:** this is the model that makes time real instead of pseudo-time, which both the physically-moving mesh (Section 1) and a genuinely periodic torque signal (Section 7) depend on.

4. Timestep Size and Inner Iterations

Setting	Value	Basis
Time-Step (Solvers > Implicit Unsteady)	5.5556×10^{-4} s	1° of rotation per timestep at 31.416 rad/s (300 rpm) — $(1/360) \times 0.2$ s per revolution. Gives exactly 360 timesteps/revolution.

Setting	Value	Basis
Maximum Inner Iterations (Stopping Criteria, top level)	10	Auto-populated by STAR-CCM+ on switching to Implicit Unsteady; kept as a reasonable starting budget for this flow's local gradients (blade tips, baffle jet impingement). To be checked against actual per-timestep residual behavior once more run data is available.

Note: Maximum Inner Iterations lives in the simulation's main Stopping Criteria folder, not nested inside the Implicit Unsteady solver node — easy to look in the wrong place.

5. Initial Condition: Warm Start

No explicit action taken — specifically, **Initialize Solution was not clicked**. Working in the Save-As copy that already carries the fully converged MRF flow field forward means the run starts from that field rather than a uniform/zero guess.

Why: the MRF solution already has the right overall flow topology (circulation loops, discharge jet, velocity magnitudes) baked in; only the periodic blade-baffle unsteadiness is missing, and that's what sliding mesh needs to develop. Warm-starting means fewer revolutions are needed to reach a stable periodic state than a cold start would need. Caveat: the first revolution or so of data reflects the solution shedding MRF-specific artifacts as much as developing real unsteadiness, so it isn't representative on its own.

6. Stopping Criteria

Criterion	Setting	Why
Maximum Physical Time	1.2 s (≈ 6 revolutions)	Primary run-length target — enough for the initial transient to wash out and several stable cycles to be captured.
Maximum Steps	5000	Generous backstop; not the controlling limit given a fixed Time-Step, just a safety ceiling.
Moment 1 Monitor Criterion	left Disabled	This Asymptotic criterion was tuned during MRF for a monotonically-converging value. Torque is now meant to oscillate periodically, so a "has it stopped changing" check is the wrong tool; convergence will instead be judged by cycle-to-cycle waveform repeatability (Section 7).

7. Torque Monitoring for a Periodic Signal

Reports > Moment 1 unchanged (Axis [0,0,1], Origin [0,0,0], on ImpellerWall). Monitors > Moment 1 Monitor: both the top-level **Trigger** and the nested **Update > Trigger** changed from **Iteration** to **Time Step**.

Why: left on Iteration, the monitor records once per inner iteration — mostly still-converging noise from within each timestep's 10-iteration loop. Time Step records exactly one value per completed timestep: the actual converged torque at that instant. With a fixed Time-Step, sample index maps directly to physical time (360 samples = 1 revolution), so periodicity can be read straight off the plot.

How Po will be extracted once running long enough: once consecutive revolutions show the same repeating waveform shape, average torque over one or more *whole* revolutions (never a partial window, which would bias the mean toward one part of the blade-baffle cycle) and feed that mean into the same Po formula used for the MRF result.

8. Solution History (for Post-Run Animation)

Setting	Value	Why
Regions recorded	RotatingFluid + StationaryFluid (both)	Only option available — this version records at Region/volume level, not a pre-selected derived part like the Plane Section; any cut is built during playback instead. Both regions are required regardless, since the point is to see flow across the whole tank (baffles included), not just the rotating zone.
Field recorded	Velocity > Lab Reference Frame > Magnitude (scalar only)	Sufficient for a colored contour animation; skips recording the full 3-component vector to cut storage roughly 3x for the same visual result.

Setting	Value	Why
Save frequency	Every 8th timestep ($\approx 8^\circ/\text{frame}$, ≈ 45 frames/revolution)	Smooth enough for visual playback without saving full-volume data at every single 1° timestep, which would be excessive given known disk constraints on this machine (see Section 9).
Compression	Lossless (default)	Kept as-is.

9. Save Discipline

1) Save-As performed before starting the run: RotorLab_SlidingMesh_v1.sim — a clean baseline with the warm-started field and every setting above configured, zero transient timesteps run. 2) Auto Save reconfigured to retain multiple numbered copies rather than overwrite a single file — directly addressing the failure mode behind the earlier corruption incident, where a single overwritten file offered no fallback when a write was interrupted. 3) Plan to take additional manual, incrementing Save-As checkpoints roughly every 500-1000 timesteps during the run, on top of Auto Save.

10. Early Run Sanity Checks (Observed So Far)

- The Iteration counter continuing from 1500 (i.e. 1501, 1502, ...) is expected — it is a lifetime counter across the whole sim file and does not reset when the time model changes. The counters that matter, Time-Step and Physical Time, both restarted cleanly: TimeStep 1 landed at Time = 5.5556×10^{-4} s, TimeStep 2 at 1.11112×10^{-3} s — exactly one and two Time-Steps, confirming Δt is being applied correctly.
- Converged (end-of-inner-iteration) torque by TimeStep 1 was -0.4204 N·m, and by TimeStep 25-27 had settled to roughly -0.4238 to -0.4244 N·m — close to the prior MRF value of -0.41666 N·m, with no divergence or blow-up. Too early (under 30 of 360 timesteps per revolution) to say anything about periodicity yet, but a stable, sane start.
- Occasional "Zero-area face(s); area adjusted for N faces" messages appeared starting around TimeStep 26, flagging 1-4 faces out of roughly 44,000 interface intersection faces. Assessed as a minor, expected numerical artifact from the interface recalculating through certain rotational alignments, not a defect — worth continuing to watch only if the flagged count grows substantially over time rather than staying at single digits.

11. Next Checkpoint

Let the run proceed to roughly TimeStep 360 (one full revolution, Time ≈ 0.2 s) before drawing any conclusions about periodicity or convergence. That is the first point with enough data to meaningfully evaluate the transient result.